SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

20/20

Annexure A

Scenario-Based

Code of Conduct:

I M.Madhulika (192478198) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M.Madhulika

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin g of the scenario and context	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

Smart campus Network Design.

Problem statement

A smart campus has been assigned the IPv4 address block 192.168.100.0/24. Three departments required separate networks with different subnet masks.

- School of computing - /25
- School of Electronics - /26
- School of Mechanical Engineering - /27

The objective is to allocate IP addresses efficiently while ensuring each department has sufficient host addresses.

Step1 : understanding the subnet Mask.

Subnet Mask	Binary Mask	Total Addresses	usable Hosts
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30

Formula:

- Total Addresses = 2^(32 - prefix length)
- usable Hosts = Total Addresses - 2

2) Company Network Expansion.

Problem statement

A software company has received the private network 10.10.0.0/16. Different departments require different subset size.

Subnet Allocation

Software Development (/24

- Network Address : 10.10.10.0
- First Host : 10.10.10.1
- Last Host : 10.10.0.254
- Broad cast Address : 10.10.0.255

Quality Assurance (/25)

- Network Address : 10.10.1.0
- First host : 10.10.1.1
- Last host : 10.10.1.126
- Broadcast Address : 10.10.1.127
- Usable Hosts : 126

Human Resources (/26)

- Network Address : 10.10.1.128
- First Host : 10.10.1.129
- Last Host : 10.10.1.190

oadcast Address : 10.10.1.191
usable Hosts : 62

Executive Management (127)

- Network Address : 10.10.1.192
- First Host : 10.10.1.193
- Last Host : 10.10.1.222
- Broad cast Address : 10.10.1.223
- usable Hosts : 30

Remaining unused Address space.

10.10.1.224 - 10.10.255.255

The remaining space can be reserved for future departments or network expansion

conclusion

The allocation subnets satisfy the requirements of each department while leaving a large number unused address poor for scalability.

③ IPv6 Addressing.

Introduction

IPV6 uses 128-bit addresses, providing an ~~enadion~~ enormous address space compared to IPv4. It suppo hierarkal addressing, auto configuration, and improved routing efficiency.

Given Address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1) compressed Address

Leading zeros are removed, and consecutive groups of zeros are replaced with "::"

compressed Address:

2001:db8:abcd::/64.

2) Expanded Address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

3) Network Prefix

Prefix length = /64

Network portion :- 2001:0db8:abcd:0000

Host portion :- 0000:0000:0000:0000.

4) Number of Hosts Addresses

Host bits = 128 - 64 = 64

Total address: 2^64 = 18446,744,073,709,551,616.

conclusion :

A /64 IPv6 subnet is the standard size and supports over 18 quintillion interface identifiers making it ideal for IoT, healthcare and enterprise deployments

outing Table lookup

Routing Table.

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP: 192.168.2.45

Routing Process

Step 1:- compare the destination IP with every routing table entry.

Step 2:- 192.168.2.45 belongs to the network 192.168.2.0/24

Step 3:- The router selects the matching route and ~~frame~~ forwards the packets through the interface connected to the 192.168.2.0/24 network.

Step 4:- If no route matches, the router forwards the packet through the default route.

Conclusion:-

Routers use the longest-prefix matching algorithm to determine the best route. If no specific route exists, the default route forwards the packet to another router or ISP

5) LAN configuration

Administration LAN

Network : 192.168.10.0/26

- Network Address : 192.168.10.0
- Broadcast Address : 192.168.10.63.
- Host Range : 192.168.10.1 – 192.168.10.62.
- Default gateway :- 192.168.10.1

Suggested computed Addresses

PC1 = 192.168.10.2

PC2 = 192.168.10.3

PC3 = 192.168.10.4

Finance LAN

Network : 192.168.10.64/26

- Network Address : 192.168.10.64
- Broadcast Address : 192.168.10.127
- Host Range : 192.168.10.65 – 192.168.10.126
- Default gateway : 192.168.10.65

ggested computer Addresses

PC1 : 192.168.10.66

PC2 : 192.168.10.67

PC3 : 192.168.10.68

communication.

Hosts in the administration LAN use 192.168.10.1 as the default gateway.

Hosts in the Finance LAN use 192.168.10.65 as the default gateway.

The layer-3 router forwards packets between the LANs, enabling communication while keeping each subnet logically separated.